

Adaptive Preference Discovery Agent



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Subject	Human Computer Interaction
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Project Topic

A. Topic

This project proposes an AI fashion recommendation agent that distinguishes between a user's relatively persistent style preferences and transient needs that arise from their current situation and continuously reconciles the two over the course of a conversation. On top of a baseline style profile that the user can review and revise, the system incorporates session-level context such as schedule, activity, location, and desired impression. The agent then determines which preferences are currently uncertain and presents either a dialogue or outfit cards depending on whether that preference is more easily expressed in language or more suited to visual comparison. By iteratively incorporating the user's natural language responses and image-based like/dislike selections, the system refines the current intent, and once it determines that sufficient information has been gathered, it transitions to a final recommendation. Through this, the project explores how user modeling that separates baseline preferences from transient context, combined with an interaction that adaptively integrates dialogue and image selection, affects the user's ability to express intent, sense of control, and overall recommendation experience.

B. Hypothesis

- **H1: Effect of separating and reconciling baseline preferences and context**

Compared to an approach that adjusts the two types of information without distinguishing between them, an approach in which users separately review and revise their baseline preferences and current context increases their sense of control and helps them make choices that satisfy both authenticity (feeling "like themselves") and situational appropriateness.

- **H2: Effect of adaptive interaction**

Compared to confirming all preferences through text-based dialogue alone, or presenting images in a fixed manner, an approach that adaptively switches between dialogue and image-based selection depending on the type of preference and the system's level of uncertainty reduces the burden of expressing one's intent and increases the degree to which users feel the AI has understood their intent.

2-1

Persona #1

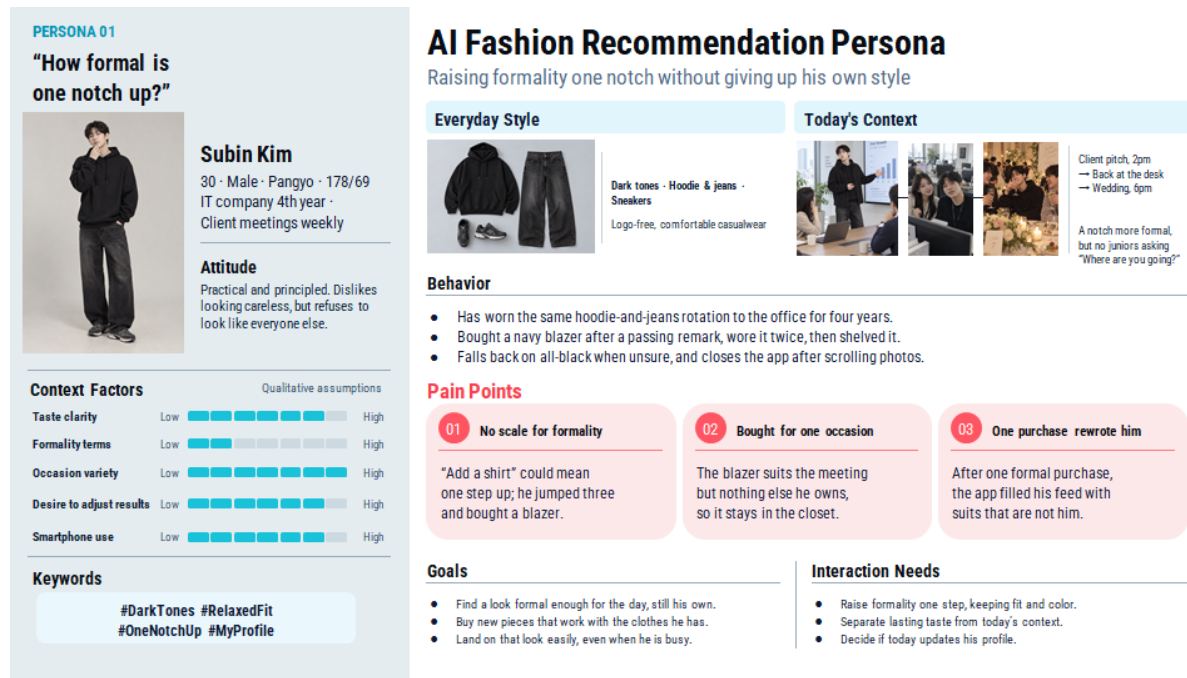


Figure1. Summary of Persona1

Description

Dark-toned clothing makes up 90% of his closet; he doesn't prefer highly saturated colors. Hoodies and jeans have been his go-to for 4 years running, and he avoids buying anything with a body-hugging fit because it feels uncomfortable. Clothes with logos, patterns, or decorative details bother him all day if he wears them. His formal wardrobe is limited to a black matching suit and a navy blazer, both purchased last month for a wedding. He's aware of trends but doesn't follow them. He mainly shops on Musinsa. As client-facing meetings become more frequent, he struggles to decide how much to adjust his usual style without feeling overdressed. He wants help comparing small changes to familiar outfits and does not want occasional purchases for formal events to redefine his everyday preferences.

2-2

Scenario #1

A. Problem Scenario

- Thursday, 5:40 PM. Subin Kim is waiting for the company shuttle home when he remembers the external client meeting scheduled for the next morning. He usually wears a black hoodie and jeans to work, and it's never really been an issue at the office. But as face-to-face meetings with clients have become more frequent lately, he's started worrying more about what to wear. He still can't shake something his team lead said with a smile after the last meeting: "Subin, maybe throw on a shirt next time."
- Subin opens Musinsa and searches "men's client meeting outfit" and "men's business casual." But most of the results — shirts, slacks, blazers — are styles he almost never wears. Ever since he bought a black set and a navy blazer last month for a wedding, the recommendation algorithm has kept showing him formal clothing as well. Looking at the recommended items, he feels like they've drifted away from his usual taste.
- At the same time, showing up in his usual hoodie feels risky given what his team lead said and the impression he wants to make on the client. On the other hand, wearing the blazer feels overly formal like something that would make his junior coworkers ask, "Hyung, what's going on with you today?" What Subin needs isn't simply "formal clothes," but a way to figure out how far to dial up the formality for today's situation while still keeping his usual style intact.
- But that's hard to put into an exact search query. As he goes back and forth comparing multiple shopping sites and outfit photos, the time left before the shuttle arrives keeps shrinking. Unable to

reach a clear decision, Subin tentatively decides to wear his usual black hoodie and jeans the next day.

B. Detailed Scenario

The following scenario revisits the same situation, showing how it could unfold if the user had access to the proposed agent.

1. Identifying the base style and the current situation

Subin launches our system. The system has a base style profile stored from Subin's earlier input:

- Prefers dark colors
- Prefers a relaxed fit
- Prefers designs with minimal logos, patterns, or decoration
- Prefers a casual style
- Values comfortable wear

Rather than modifying the base profile right away, **the agent treats it as background information for the current conversation.**

AI asks: "What situation are you dressing for?"

Subin types: "I have a client meeting tomorrow. I don't really want to wear my usual hoodie, but I don't want to dress like I'm in a suit either."

The system extracts the following as the current session's context:

- Purpose: Client meeting,
- Desired impression: Higher level of formality than usual
- Impression to avoid: Excessive formality
- Baseline preferences to preserve: Prefers a relaxed fit

For information that needs factual confirmation, like "don't want to wear hoodie," the agent asks a short text-based question.

AI asks: "Would you prefer not to wear a hoodie at all tomorrow, or are you mainly trying to look a little more polished than usual?"

Subin replies: "I'd rather not wear a hoodie tomorrow. I still want to be comfortable, but I want to look a bit more put together."

The system records "avoid a hoodie," "maintain comfort," and "slightly increase formality" as temporary requirements for this session. It does not revise Subin's baseline preference for casual clothing.

On the other hand, the agent judges that the exact level and style of formality Subin wants are difficult to infer from language alone.

2. Exploring uncertain preferences through images

The agent therefore decides that comparing outfit images will capture Subin's intent more effectively than asking him to rate his desired level of formality on a numerical scale.

Two outfit options are shown on screen

Outfit A: Black knit + relaxed black pants + minimal sneakers

Outfit B: White shirt + navy blazer + slacks + loafers

Rather than simply asking "Which outfit do you like better?", the agent frames the question to explicitly include the current situation.

AI asks: "If you were heading to tomorrow's client meeting, which of these would you lean toward wearing?"

Subin selects Like for Outfit A.

However, since Outfit A and Outfit B differ in several respects such as color, fit, outerwear, and shoes the system doesn't simply interpret every feature of A as Subin's preference. Judging that there's significant uncertainty about the reason behind his choice, it presents a short follow-up question.

AI asks: "What's the main reason you preferred Outfit A?"

- It looks comfortable
- It's less formal
- I like the color

- I like the overall vibe

Subin selects "It's less formal."

The system reflects this not as a long-term baseline preference, but as a requirement specific to the current client-meeting session.

3. Confirming necessary information through dialogue

Next, the system determines that while the amount of travel and physical activity involved matters for the current situation, this is information better gathered by asking directly rather than by showing images.

AI asks: "Will you be doing a lot of moving around or walking tomorrow?"

Subin replies: "I'll go from Pangyo to Gangnam and then head back to the office after the meeting. I think I'll be walking quite a bit."

The system adds activity level, extended wear, and maintaining comfort to the current context.

It then summarizes what it has gathered so far and checks it with Subin.

AI says: "It sounds like you want to keep your usual comfortable, dark-toned style, but look a notch more polished than a hoodie without feeling like you're in a suit."

Subin replies: "That's right."

4. Fine-tuning through visual comparison

The system has now sufficiently identified the key situational factors and Subin's baseline preferences, but it's still uncertain about exactly which way of raising the formality level feels most natural to him.

This time, it presents two outfits that are kept as similar as possible across other attributes, varying the styling of the top.

Outfit A: Charcoal crewneck knit + black pants + gray sneakers

Outfit B: Dark navy overshirt + black pants + gray sneakers

Subin selects Outfit B.

From this, the system infers that Subin isn't rejecting a neater style itself. Rather, he finds formality uncomfortable when it breaks sharply from his usual style, as with a blazer or suit.

The agent checks this interpretation again.

AI says: "It seems like you'd rather raise the level of neatness in a way that flows naturally from your usual style like an overshirt rather than something like a blazer."

Subin confirms this.

After the user confirms the interpretation, the agent judges that the remaining uncertainty is unlikely to justify further questions and proceeds to recommendations.

5. Final recommendation and rationale

Based on Subin's baseline preferences and the situational requirements identified so far, the system proposes three outfit options.

The first outfit keeps the most elements of Subin's usual style; the second strikes a balance between his usual style and the demands of the meeting; and the third stays reasonably close to his existing taste while allowing him to try something a bit newer.

Subin selects the second outfit because it feels appropriate for the client meeting and his preference.

Along with the recommendation, **the system distinguishes which elements were preserved from his existing preferences and which were adjusted for today's situation.**

Elements preserved from his usual preferences

- Dark colors
- Relaxed silhouette
- Design with minimal logos and decoration
- Comfort and ease of movement

Elements adjusted for the current situation

- Comfortable sneakers to accommodate the expected walking
- An overshirt in place of a hoodie

- A higher level of formality than usual
- A neater silhouette suited to the client meeting

Elements clarified through this conversation

- Blazer-level formality feels like too much
- A semi-casual style that connects naturally with his usual look is acceptable

The agent asks: "You preferred an overshirt to a blazer for tomorrow's client meeting. Should I remember this preference for future recommendations, or apply it only to this occasion?" Subin replies: "Just this occasion. For everyday outfits, I still prefer hoodies and casual clothes." The system records his preference for an overshirt as specific to the current client-meeting session and leaves his baseline style profile unchanged.

Subin judges that with this recommended outfit, he won't come across as overly casual to the client, nor will he look out of place or awkward to his coworkers.

3-1

Persona #2

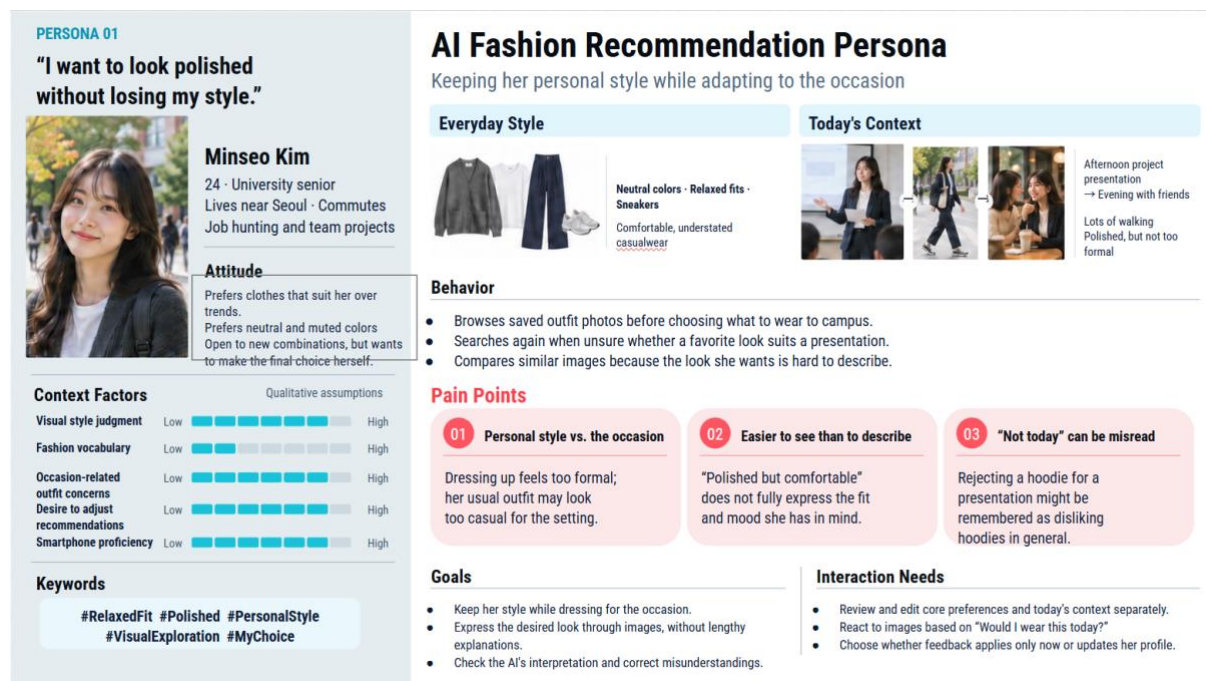


Figure2. Summary of Persona #2

Description

Minseo Kim is a 24-year-old university senior balancing job hunting and team projects. She prefers neutral colors, relaxed fits, and sneakers, valuing comfortable clothes that suit her rather than following trends. Before leaving for campus, she browses saved outfit photos but struggles to balance a polished appearance with her personal style on days that involve both a presentation and meeting friends. Although she can recognize the look she wants in images, describing it through fashion terms or detailed sentences feels difficult, so she repeatedly compares similar photos. She also does not want to reject an outfit for a presentation to be interpreted as a lasting dislike of that style. She therefore wants to adjust her everyday preferences and the day's context separately, refining her choices by reacting to outfit images. She wants to compare outfit images, check whether the recommendations reflect her intentions, and decide which preferences should carry over to future occasions.

3-2

Scenario #2

A. Problem scenario

- Monday afternoon, Minseo has a team project presentation at school the next day. Since the weather has turned cooler, she's made evening plans with friends in Seorae Village afterward browsing several small shops, taking photos, and walking around, so it won't be a short distance to cover. She usually likes a combination of a loose-fitting knit, wide pants, and sneakers, but for the presentation she doesn't want to look too relaxed or unpolished, since that might make her seem unprepared.
- Minseo first picks out a few outfit photos she's saved on Ably and tells Gemini "I have a project presentation tomorrow and I want to look neat without looking too much like I'm in a suit. I'm also meeting friends afterward." Gemini suggests a generic combination that looks presentation-appropriate — blazer, shirt, slacks. Minseo doesn't think it's a wrong answer exactly, but it feels far from the style she normally wears. She tries to explain more specifically that she wants to "look neat without losing my own style," but struggles to find the right words for it.
- She then searches Zigzag for "women's college presentation look" and "project presentation outfit." Filters and styled content bring up plenty of items, but the results are full of things like blouses, jackets, and loafers that seem tailored only to the presentation setting.
- On Ably, she also browses the "work outfit" and "semi-formal" categories, but she must keep manually balancing, on her own, the line between the comfortable casual items she's liked before and the neatness she needs today. Still, she can't keep spending this much time hunting for something she likes, since she also needs to do a final check of her presentation prep tomorrow so she ends up having to settle for a compromise. Situations like this have been happening more often lately, and it's been a real source of stress. Even though these are apps she uses all the time, she feels frustrated that there's no feature that can account for "my personal taste," "presentation formality," and "plans with friends" all at once.
- Minseo finds herself torn between the following:
 - Sticking with her usual style might make her look too casual for the presentation.
 - Choosing a jacket and loafers would feel unlike her and overdressed.
 - Factoring in the evening plans with friends makes it hard to decide based on the presentation alone.
 - She worries that telling Gemini "I don't want a blazer today" might be interpreted as disliking blazers altogether going forward.
- The next day, Minseo ends up wearing her familiar knit and an ordinary pair of pants, but she keeps second-guessing her outfit throughout the presentation. After somehow getting through it, she arrives at the meetup spot with her friends only to feel her mood sour a bit at the sight of her friends in stylish, trendy outfits that fit the Seorae Village vibe, making her own outfit feel a little underwhelming by comparison.
- On her way home, Minseo feels the need for an app that does more than just list products — one that could help her figure out, in situations like today's presentation, evening plans, and amount of walking involved, which elements to adjust and by how much while keeping her usual style intact.

B. Detailed scenario

The following scenario revisits the same situation, showing how it could unfold if the user had access to the proposed agent.

1. Identifying the base style and the current situation

Minseo launches our system. The system has a base style profile stored from Minseo's earlier input:

- Prefers a relaxed fit
- Prefers sneakers
- Prefers a comfortable but neat casual style
- Dislikes clothes with excessive logos/decoration
- Tends to judge by comparing images
- Wants to make the final choice herself

Rather than modifying the baseline profile right away, **the agent treats it as background information for the current conversation.**

AI asks: "What situation are you dressing for?"

Minseo types: "I have a project presentation at school tomorrow. My usual style feels too casual and unpolished for that... After the presentation, I have dinner plans with friends in Seorae Village — I'll have to walk a lot, and I want to keep my usual style while looking a bit neater."

The system extracts the following as the current session's context:

- Purpose: Team project presentation
- Following schedule: Dinner plans with friends
- Constraint: Extended travel and walking
- Desired impression: Neatness, looking prepared
- Impression to avoid: Usual overly casual look
- Baseline preferences to preserve: Comfortable fit, neutral colors, sneakers

For information that needs factual confirmation, like "having to walk a lot," the agent asks a short text-based question.

AI asks: "Will it be hard to change shoes after the presentation, and will you be on the move continuously?"

Minseo replies: "Yeah, I want to wear sneakers all day"

The system reflects activity level and sneakers as today's essential constraints.

On the other hand, it judges that "neat, but still like me" is difficult to interpret from language alone.

2. Exploring uncertain preferences through images

The system determines that comparing actual outfits will capture Minseo's intent better than an additional text-based question like "Please rate the level of casualness you want from 1 to 5."

Two outfit options are shown on screen

Outfit A: Black blazer + shirt + wide-leg slacks + brown sneakers

Outfit B: Charcoal knit + wide-leg slacks + white sneakers

Rather than simply asking "Which outfit do you like better?", the agent frames the question to explicitly include the current situation.

AI asks: "Considering both tomorrow's presentation and the dinner plans, which look would you lean toward?"

Minseo selects Like for Outfit B.

However, since Outfit A and Outfit B differ in several respects such as type, color, fit, outerwear, and shoes the agent doesn't simply interpret every feature of B as Minseo's preference. Judging that there's significant uncertainty about the reason behind her choice, it presents a short follow-up question.

AI asks: "What's the main reason you picked B?"

- It looks comfortable
- It's less formal
- I like the color
- I like the overall vibe

Minseo selects "It looks comfortable"

At this point, the agent checks its interpretation.

AI says: "It sounds like you want to keep the relaxed fit and sneakers while choosing a slightly more polished top for the presentation."

Minseo confirms this.

The system records it as a requirement for the current presentation-and-dinner session. The agent then checks what contextual information is still missing.

3. Confirming necessary information through dialogue

Based on the preceding context and choices, the agent determines that image selection alone has limits for capturing the shifting context, and that it needs to ask an additional direct question.

The agent asks: "Would you like to wear the same outfit for both plans, or would you be comfortable making a small change after the presentation?"

Minseo replies: "I'd rather keep the same outfit. I won't have time to go home and change."

The system records that the outfit needs to work for both the presentation and dinner without requiring a change of clothes. It then confirms what it has gathered so far with Minseo.

AI says: "It sounds like you want to keep your usual casual style but lean it toward the more formal end within that range."

Minseo replies: "That's right."

4. Fine-tuning through visual comparison

The agent has identified Minseo's baseline preferences and the main requirements for tomorrow but remains uncertain about how she would like to make her outfit look more polished.

The agent decides that another visual comparison would help clarify this preference. The system displays two outfits with the same trousers and sneakers, allowing Minseo to compare two approaches to styling the top.

Outfit A: Dark gray crewneck knit + wide-leg black trousers + white sneakers

Outfit B: Dark gray Minimal overshirt + ivory inner T-shirt + wide-leg black trousers + black sneakers

Minseo selects Outfit B.

The agent tentatively interprets her choice as a preference for adding structure through an overshirt while keeping the rest of the outfit relaxed. It then checks this interpretation.

AI says: "It seems like you would prefer to add a little structure with an overshirt while keeping the relaxed trousers and sneakers"

Minseo confirms this.

After Minseo confirms this interpretation, the agent judges that the remaining uncertainty is unlikely to justify further questions and proceeds to recommendations.

5. Final recommendation and rationale

Based on Minseo's baseline preferences and the situational requirements identified so far, the system presents three outfit options using product catalogs from Zigzag and Ably and pre-built outfit combinations.

The first outfit preserves the most elements of Minseo's usual style. The second balances her everyday preferences with the formality required for the presentation and her plans afterward. And the third stays within her preferred neutral palette while introducing a slightly more distinctive style. Minseo selects the second outfit because it feels appropriate for the presentation while remaining comfortable enough for her evening plans.

Along with the recommendation, **the system distinguishes which elements were preserved from her existing preferences and which were adjusted for today's situation.**

Elements preserved from her usual preferences

- Relaxed silhouette
- Neutral and low-saturation colors
- Minimal logos and decorative elements
- Comfortable sneakers

Elements adjusted for the current situation

- A neater upper-body silhouette for the presentation
- A slightly higher level of formality than usual

- Styling that can transition naturally from the presentation to dinner plans

Elements clarified through this conversation

- Overshirt-based outfit that feels polished without sacrificing comfort in this occasion
- She prefers to maintain a relaxed silhouette even when dressing more formally

The agent asks: "You chose an overshirt to look polished while staying comfortable for tomorrow's presentation and dinner plans. Should I also use this preference in future recommendations, or apply it only to this occasion?"

Minseo replies: "Only this occasion. I don't necessarily want my everyday outfits to look this polished." The system records this preference as specific to the current presentation-and-dinner session and leaves her baseline style profile unchanged.

Minseo feels that the selected outfit will help her look prepared for the presentation while still feeling like herself. She also feels that the outfit is comfortable and casual enough to continue wearing during the dinner plans afterward.

4

Claim Analysis

A. Claims from the problem scenarios

Situational / system attribute	HCI factor	Claim analysis
P1. Purchase-history personalization	Usefulness Emotion	(+) Recent purchases can surface related formal items for another occasion. (-) A one-off wedding purchase can dominate later suggestions, making Subin feel that his usual casual taste is no longer represented.
P2. Occasion-keyword search	Usefulness Usability	(+) Occasion keywords quickly provide examples of meeting-appropriate clothing. (-) Broad results do not express how much formality Subin will accept; comparing sites and photos consumes his limited time and leaves the decision unresolved.
P3. Text-based outfit consultation	Usability Usefulness	(+) Natural-language input lets Minseo state the presentation and evening plans together. (-) The generic blazer combination leaves her struggling to explain the visual difference between looking neat and feeling unlike herself.
P4. Occasion-based shopping categories	Usefulness Emotion	(+) Categories offer recognizable examples of presentation-appropriate outfits. (-) Presentation-focused results leave Minseo to reconcile her usual style, evening plans, and walking needs herself; her eventual compromise leaves her dissatisfied.

B. Claims from the detailed scenarios

Situational / system attribute	HCI factor	Claim analysis
D1. Baseline / session separation	Usefulness Emotion	(+) Keeping meeting-specific and presentation-specific feedback in the current session can preserve both users' everyday preferences while accommodating tomorrow's needs. (-) If a situational preference later becomes persistent, session-only treatment may miss that change and require the user to express it again.
D2. Adaptive dialogue and image comparison	Usability Usefulness	(+) Text captures concrete walking and schedule constraints, while outfit images help both users express a desired level of neatness without precise fashion vocabulary. (-) An inappropriate switch or too many alternating questions can interrupt comparison and add effort before either user reaches a decision.
D3. Reason clarification after an image choice	Usefulness Usability	(+) Asking why A was chosen reduces the chance of attributing the choice to every visible feature; the same image preference can reveal different priorities for each user. (-) Another question increases interaction burden, and predefined reasons can steer the response if they do not capture what the user actually noticed.
D4. Multiple-context constraint checking	Usefulness Usability	(+) Explicitly checking mobility and later activities helps the recommendation address the whole day rather than the formal event alone. (-) Reconfirming context already stated can feel repetitive; balancing several demands may still produce a compromise that neither event fully satisfies.
D5. Focused visual comparison	Usability Usefulness	(+) Keeping shared outfit elements stable helps each user compare a smaller change and indicate an acceptable way to look neater. (-) The tops still differ in visual attributes beyond formality; choosing an overshirt does not establish which individual attribute caused the preference.
D6. Confirmation of AI interpretation	Usefulness Emotion	(+) Restating the inferred balance between comfort and neatness lets users check whether the agent understood them before the recommendation. (-) A leading summary can encourage agreement even when an interpretation is only partly correct; repeated confirmations can also become tiring.
D7. Three alternatives with an explained rationale	Usefulness Emotion	(+) Comparing alternatives and their rationale can help users judge the trade-off between personal style and occasion suitability while retaining the final choice. (-) Three curated options can narrow exploration; a persuasive explanation may increase confidence even though actual comfort and others' reactions remain unverified.

